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Aim:

To design library management system.

Description:

The webpage is created using various HTML and CSS tags to display book information in a structured table. JavaScript is used to fetch data from an XML file, dynamically generate HTML content, and interact with the webpage.

XML Concepts Used:

- **Structured Data Storage –**
XML is used to store book records in a structured, hierarchical format. Each `<book>` element contains details about a book.
- **Elements –**
Each book record has child elements such as `<id>`, `<title>`, `<author>`, `<category>`, `<price>`, and `<status>` to store respective information.
- **Hierarchical Organization –**
All `<book>` elements are wrapped inside a `<library>` root element, making it easy to manage and parse multiple book records.
- **Data Retrieval –**
JavaScript can fetch and parse XML using `XMLHttpRequest` to access each element's content dynamically.
- **Extensibility –**
New books can be added to the XML file without changing the structure, making it flexible for future expansion.

JavaScript Concepts Used:

- **AJAX (XMLHttpRequest) –**
Used to fetch data from the XML file without reloading the webpage.

Code:

Index.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Library Management </title>
</head>
<link rel="stylesheet" href="style.css">
<style>
  body{
    background-color: rgb(172, 189, 126);
  }
```

```
</style>
<body>

<h1 class="title">Library Management System</h1>
<hr>
<nav class="container">

  <h3> <a href="view.html"> View Books  </a></h3>
  <h3> <a href="add.html"> Add Books  </a></h3>
</nav>
<marquee behavior="scroll" direction="left" scrollamount="40">
  
  
  
  
  
</marquee>
<hr>
</body>
<footer>
  <div class="footer-content">
    <p><b>Contact Us:</b></p>
    <p>Email: library@example.com</p>
    <p>Phone: +1 234 567 890</p>
    <p>Address: 123 Library Street, Booktown</p>
  </div>
</footer>
```

View.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>View Books</title>
<link rel="stylesheet" href="style.css">
</head>
<style>
  body{
    background-color: rgb(172, 189, 126);
  }
</style>
<body>
<h1 class="title">Library Management System</h1>
<hr>
<table id="output"></table>
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

```
</html>
```

Add.html:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Add Books</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<style>
```

```
  body{
```

```
    background-color: rgb(172, 189, 126);
```

```
  }
```

```
</style>
```

```
<body>
```

```
  <h1 class="title">Library Management System</h1>
```

```
<hr>
```

```
  <form method="post" action="add.html" class="form">
```

```
    <h2>Add Book</h2>
```

```
    <label for="id">ID:</label><br>
```

```
    <input type="text" id="id"><br><br>
```

```
    <label for="title">Title:</label><br>
```

```
    <input type="text" name="title" id="title"><br><br>
```

```
    <label for="author">Author:</label><br>
```

```
    <input type="text" name="author" id="author"><br><br>
```

```
    <label for="category">Category:</label><br>
```

```
    <input type="text" name="category" id="category"><br><br>
```

```
    <label for="price">Price:</label><br>
```

```
    <input type="number" name="price" id="price"><br><br>
```

```
    <label for="status">Status:</label><br>
```

```
    <input type="text" name="status" id="status"><br><br>
```

```
    <input type="submit" value="Add" onclick="add()">
```

```
</form>
```

```
<script src="script.js"></script>
```

```
<hr>
```

```
</body>
```

```
</html>
```

Library.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<library>
```

```
  <book>
```

```
    <id>101</id>
```

```
    <title>Clean Code</title>
```

```
<author>Robert C. Martin</author>
<category>Programming</category>
<price>$450</price>
<status>Available</status>
</book>
<book>
  <id>102</id>
  <title>Introduction to Algorithms</title>
  <author>Thomas H. Cormen</author>
  <category>Computer Science</category>
  <price>$850</price>
  <status>Issued</status>
</book>
<book>
  <id>103</id>
  <title>Design Patterns</title>
  <author>Erich Gamma</author>
  <category>Software Engineering</category>
  <price>$650</price>
  <status>Available</status>
</book>
<book>
  <id>104</id>
  <title>Operating System Concepts</title>
  <author>Silberschatz</author>
  <category>Operating Systems</category>
  <price>$720</price>
  <status>Issued</status>
</book>
<book>
  <id>105</id>
  <title>Computer Networks</title>
  <author>Andrew S. Tanenbaum</author>
  <category>Networking</category>
  <price>$680</price>
  <status>Available</status>
</book>
<book>
  <id>106</id>
  <title>Database System Concepts</title>
  <author>Abraham Silberschatz</author>
  <category>Database</category>
  <price>$700</price>
  <status>Issued</status>
</book>
<book>
```

```
<id>107</id>
<title>Java: The Complete Reference</title>
<author>Herbert Schildt</author>
<category>Programming</category>
<price>$550</price>
<status>Available</status>
</book>
<book>
  <id>108</id>
  <title>Python Crash Course</title>
  <author>Eric Matthes</author>
  <category>Programming</category>
  <price>$480</price>
  <status>Available</status>
</book>
<book>
  <id>109</id>
  <title>Artificial Intelligence</title>
  <author>Stuart Russell</author>
  <category>AI</category>
  <price>$900</price>
  <status>Issued</status>
</book>
<book>
  <id>110</id>
  <title>Machine Learning</title>
  <author>Tom Mitchell</author>
  <category>AI</category>
  <price>$820</price>
  <status>Available</status>
</book>
<book>
  <id>111</id>
  <title>Web Technologies</title>
  <author>Ullman</author>
  <category>Web Development</category>
  <price>$400</price>
  <status>Available</status>
</book>
<book>
  <id>112</id>
  <title>Software Testing</title>
  <author>Ron Patton</author>
  <category>Software Engineering</category>
  <price>$460</price>
  <status>Issued</status>
```

```

</book>
<book>
  <id>113</id>
  <title>Cloud Computing</title>
  <author>Rajkumar Buyya</author>
  <category>Cloud</category>
  <price>$780</price>
  <status>Available</status>
</book>
<book>
  <id>114</id>
  <title>Cyber Security Essentials</title>
  <author>Charles J. Brooks</author>
  <category>Security</category>
  <price>$620</price>
  <status>Available</status>
</book>
<book>
  <id>115</id>
  <title>Data Science Handbook</title>
  <author>Jake VanderPlas</author>
  <category>Data Science</category>
  <price>$860</price>
  <status>Issued</status>
</book>
</library>

```

Script.js:

```

const xhttp = new XMLHttpRequest();

xhttp.onload = function () {
  const xmlDoc = this.responseXML;
  const books = xmlDoc.getElementsByTagName("book");
  let output = `
<table border="1" cellspacing="0" cellpadding="8" class='tbl'>
  <thead>
    <tr>
      <th>ID</th>
      <th>Title</th>
      <th>Author</th>
      <th>Category</th>
      <th>Price</th>
      <th>Status</th>
    </tr>
  </thead>
  <tbody>
  `;
};

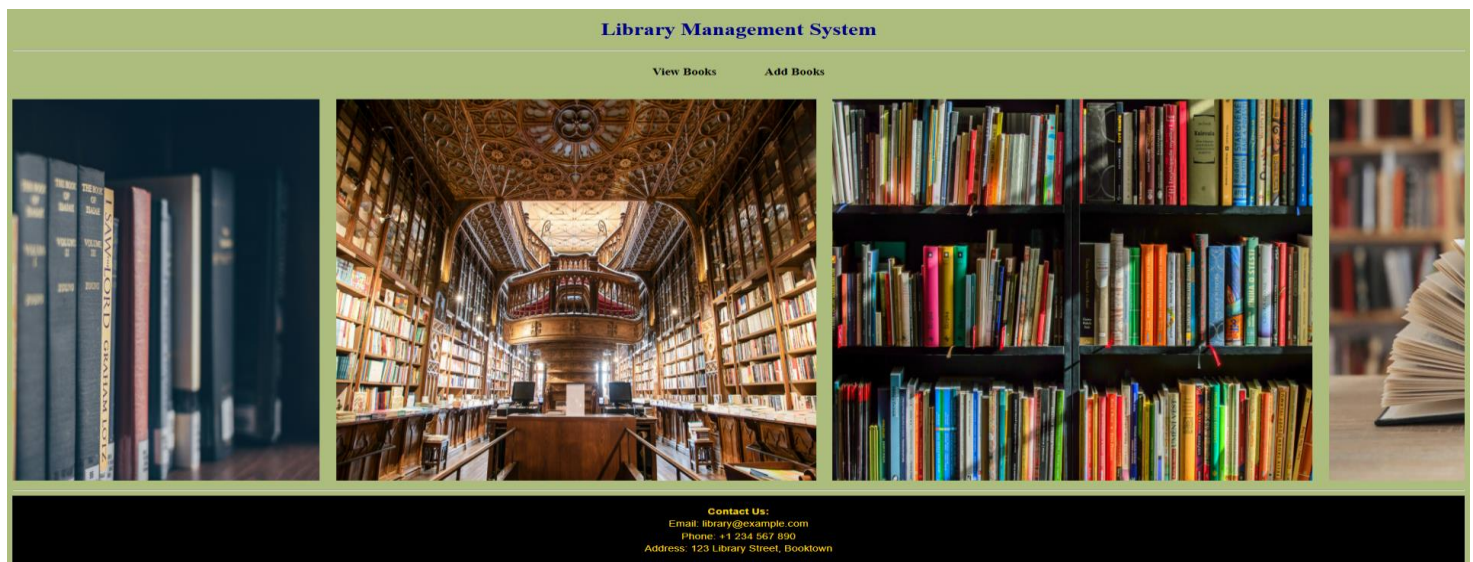
```

```

for (let i = 0; i < books.length; i++) {
  const id = books[i].getElementsByTagName("id")[0].textContent;
  const title = books[i].getElementsByTagName("title")[0].textContent;
  const author = books[i].getElementsByTagName("author")[0].textContent;
  const category = books[i].getElementsByTagName("category")[0].textContent;
  const price = books[i].getElementsByTagName("price")[0].textContent;
  const status = books[i].getElementsByTagName("status")[0].textContent;
  output += `
    <tr>
      <td>${id}</td>
      <td>${title}</td>
      <td>${author}</td>
      <td>${category}</td>
      <td>${price}</td>
      <td>${status}</td>
    </tr>
  `;
}
output += `
  </tbody>
</table>
`;
document.getElementById("output").innerHTML = output;
};
xhttp.open("GET", "library.xml", true);
xhttp.send();
function add(){
  alert("Book Added Successfully...");
}

```

Output:



Figure(a)

127.0.0.1:5500/view.html

Library Management System

ID	Title	Author	Category	Price	Status
101	Clean Code	Robert C. Martin	Programming	\$450	Available
102	Introduction to Algorithms	Thomas H. Cormen	Computer Science	\$850	Issued
103	Design Patterns	Erich Gamma	Software Engineering	\$650	Available
104	Operating System Concepts	Silberschatz	Operating Systems	\$720	Issued
105	Computer Networks	Andrew S. Tanenbaum	Networking	\$680	Available
106	Database System Concepts	Abraham Silberschatz	Database	\$700	Issued
107	Java: The Complete Reference	Herbert Schildt	Programming	\$550	Available
108	Python Crash Course	Eric Matthes	Programming	\$480	Available
109	Artificial Intelligence	Stuart Russell	AI	\$900	Issued
110	Machine Learning	Tom Mitchell	AI	\$820	Available
111	Web Technologies	Ullman	Web Development	\$400	Available
112	Software Testing	Ron Patton	Software Engineering	\$460	Issued
113	Cloud Computing	Rajkumar Buyya	Cloud	\$780	Available
114	Cyber Security Essentials	Charles J. Brooks	Security	\$620	Available
115	Data Science Handbook	Jake VanderPlas	Data Science	\$860	Issued

Figure(b)

127.0.0.1:5500/add.html

Library Management System

Add Book

ID:

Title:

Author:

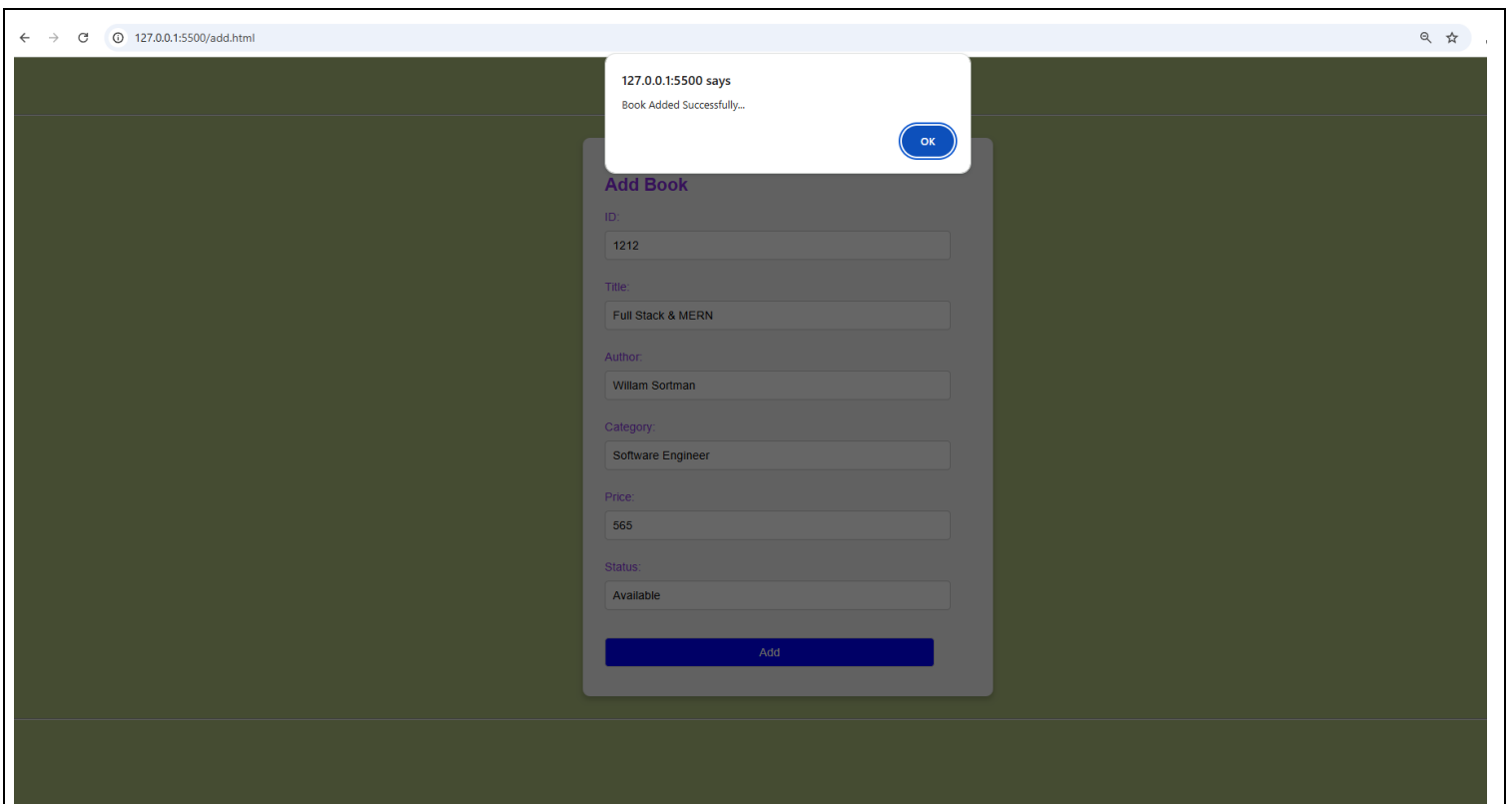
Category:

Price:

Status:

Add

Figure(c)



Figure(d)

Result:

Thus, the library management system is done and the output is taken successfully by performing add, and display operations.